武器系统动力学建模与仿真

刚体的空间运动 -V

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刚体上任意点的平动

刚体上任意点 P 在全局惯性坐标系中的位置/速度/加速度列阵:

$$
\begin{aligned}
\boldsymbol{r}_{OP} &= \boldsymbol{r}_{OR} + \boldsymbol{r}_{RP}, \\
&= \boldsymbol{r}_{OR} + \boldsymbol{A}_{OR}\boldsymbol{l}_{RP}.
\end{aligned}
$$

$$
\begin{aligned}
\dot{\boldsymbol{r}}_{OP} &= \dot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR|R}\boldsymbol{l}_{RP}, \\
&=: \dot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP}, \\
&= \dot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\boldsymbol{\omega}_{OR}.
\end{aligned}
$$

$$
\ddot{\boldsymbol{r}}_{OP} = \ddot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP}.
$$

虚速度与虚功率方程

刚体上任意点 P 的虚速度和加速度可表示为

$$\delta\dot{\boldsymbol{r}}_{OP} = \delta\dot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\delta\boldsymbol{\omega}_{OR},$$
$$\ddot{\boldsymbol{r}}_{OP} = \ddot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR} + \boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP}.$$

对于点 P 所在的微元 k, 牛顿第二运动定律给出

$$m_k\ddot{\boldsymbol{r}}_{OP} - \boldsymbol{f}_{In,k} - \boldsymbol{f}_{Ex,k} = \boldsymbol{0}.$$

上式与点 P 的虚速度的内积满足

$$\delta\dot{\boldsymbol{r}}_{OP}^{\mathrm{T}}(m_k\ddot{\boldsymbol{r}}_{OP} - \boldsymbol{f}_{In,k} - \boldsymbol{f}_{Ex,k}) = 0.$$

对刚体上所有微元求和可得

$$\sum_{k=1}^{N}\delta\dot{\boldsymbol{r}}_{OP}^{\mathrm{T}}(m_k\ddot{\boldsymbol{r}}_{OP} - \boldsymbol{f}_{In,k} - \boldsymbol{f}_{Ex,k}) = 0.$$

刚体微元间相互作用力虚功之和为零

虚功率方程:

$$\sum_{k=1}^{N} \delta \dot{\boldsymbol{r}}_{OP}^{\mathrm{T}}(m_k \ddot{\boldsymbol{r}}_{OP} - \boldsymbol{f}_{In,k} - \boldsymbol{f}_{Ex,k}) = 0.$$

对于刚体任意两微元间的相互作用力，有

$$\begin{aligned}
\delta \dot{\boldsymbol{r}}_{OP_1} &= \delta \dot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP_1}^{\mathrm{T}} \delta \boldsymbol{\omega}_{OR}, \\
\delta \dot{\boldsymbol{r}}_{OP_2} &= \delta \dot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP_2}^{\mathrm{T}} \delta \boldsymbol{\omega} R_{OR}, \\
\delta \dot{\boldsymbol{r}}_{OP_1}^{\mathrm{T}} \boldsymbol{f}12 + \delta \dot{\boldsymbol{r}}_{OP_2}^{\mathrm{T}} \boldsymbol{f}21 &= \delta \dot{\boldsymbol{r}}_{OP_1}^{\mathrm{T}} \boldsymbol{f}12 - \delta \dot{\boldsymbol{r}}_{OP_2}^{\mathrm{T}} \boldsymbol{f}12 \\
&= \delta \boldsymbol{\omega}_{OR}^{\mathrm{T}} \tilde{\boldsymbol{l}}_{P_2P_1} \boldsymbol{A}_{OR}^{\mathrm{T}} \boldsymbol{f}12 \\
&= \delta \boldsymbol{\omega}_{OR}^{\mathrm{T}} \boldsymbol{A}_{OR}^{\mathrm{T}} \tilde{\boldsymbol{r}}_{P_2P_1} \boldsymbol{f}12 \\
&= 0.
\end{aligned}$$

虚功率方程

考虑到刚体元件内力虚功率之和为零，其虚功率方程可简化为

$$\sum_{k=1}^{N} \delta \dot{\boldsymbol{r}}_{OP}^{\mathrm{T}} (m_k \ddot{\boldsymbol{r}}_{OP} - \boldsymbol{f}_{Ex,k}) = 0,$$

式中

$$\delta \dot{\boldsymbol{r}}_{OP} = \delta \dot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \delta \boldsymbol{\omega}_{OR},$$

$$\ddot{\boldsymbol{r}}_{OP} = \ddot{\boldsymbol{r}}_{OR} + \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \dot{\boldsymbol{\omega}}_{OR} + \boldsymbol{A}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \boldsymbol{l}_{RP}.$$

将虚速度和加速度表达式代入虚功率方程可得

$$0 = \sum_{k=1}^{N} \delta \dot{\boldsymbol{r}}_{OR}^{\mathrm{T}} (m_k \ddot{\boldsymbol{r}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \dot{\boldsymbol{\omega}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) +$$

$$\delta \boldsymbol{\omega}_{OR}^{\mathrm{T}} \tilde{\boldsymbol{l}}_{RP} \boldsymbol{A}_{OR}^{\mathrm{T}} (m_k \ddot{\boldsymbol{r}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \dot{\boldsymbol{\omega}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}).$$

虚功率方程的矩阵形式

刚体元件虚功率方程

$$0 = \sum_{k=1}^{N} \delta \dot{\boldsymbol{r}}_{OR}^{\mathrm{T}} (m_k \ddot{\boldsymbol{r}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \dot{\boldsymbol{\omega}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) +$$
$$\delta \boldsymbol{\omega}_{OR}^{\mathrm{T}} \tilde{\boldsymbol{l}}_{RP} \boldsymbol{A}_{OR}^{\mathrm{T}} (m_k \ddot{\boldsymbol{r}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \dot{\boldsymbol{\omega}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k})$$

可记为如下矩阵相乘的形式

$$0 =$$
$$\begin{bmatrix} \delta \dot{\boldsymbol{r}}_{OR}^{\mathrm{T}} & \delta \boldsymbol{\omega}_{OR}^{\mathrm{T}} \end{bmatrix}$$
$$\begin{bmatrix} \sum_{k=1}^{N} (m_k \ddot{\boldsymbol{r}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \dot{\boldsymbol{\omega}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) \\ \sum_{k=1}^{N} \tilde{\boldsymbol{l}}_{RP} \boldsymbol{A}_{OR}^{\mathrm{T}} (m_k \ddot{\boldsymbol{r}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}} \dot{\boldsymbol{\omega}}_{OR} + m_k \boldsymbol{A}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \tilde{\boldsymbol{\omega}}_{OR} \boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) \end{bmatrix}.$$

刚体的空间运动 -V

1 刚体元件运动学方程

2 虚功率原理

3 刚体元件动力学方程

刚体元件动力学方程

矩阵形式的虚功率方程

$$0 = \begin{bmatrix} \delta\dot{\boldsymbol{r}}_{OR}^{\mathrm{T}} & \delta\boldsymbol{\omega}_{OR}^{\mathrm{T}} \end{bmatrix} \begin{bmatrix} \sum_{k=1}^{N}(m_k\ddot{\boldsymbol{r}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) \\ \sum_{k=1}^{N}\tilde{\boldsymbol{l}}_{RP}\boldsymbol{A}_{OR}^{\mathrm{T}}(m_k\ddot{\boldsymbol{r}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) \end{bmatrix}.$$

结合独立变分原理，由上式可得刚体元件运动微分方程

$$\boldsymbol{0} = \begin{bmatrix} \sum_{k}(m_k\ddot{\boldsymbol{r}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) \\ \sum_{k}\tilde{\boldsymbol{l}}_{RP}\boldsymbol{A}_{OR}^{\mathrm{T}}(m_k\ddot{\boldsymbol{r}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR} + m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP} - \boldsymbol{f}_{Ex,k}) \end{bmatrix}.$$

刚体元件动力学方程

刚体元件运动微分方程

$$\mathbf{0}=\begin{bmatrix}\sum_k(m_k\ddot{\boldsymbol{r}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP}-\boldsymbol{f}_{Ex,k})\\ \sum_k\tilde{\boldsymbol{l}}_{RP}\boldsymbol{A}_{OR}^{\mathrm{T}}(m_k\ddot{\boldsymbol{r}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP}-\boldsymbol{f}_{Ex,k})\end{bmatrix},$$

可进一步化简为

$$\mathbf{0}=\begin{bmatrix}m\ddot{\boldsymbol{r}}_{OR}+m\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RC}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR}+m\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RC}-\boldsymbol{f}_{Ex}\\ m\tilde{\boldsymbol{l}}_{RC}\boldsymbol{A}_{OR}^{\mathrm{T}}\ddot{\boldsymbol{r}}_{OR}+\boldsymbol{J}_{R|R}\dot{\boldsymbol{\omega}}_{OR}+\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{J}_{R|R}\boldsymbol{\omega}_{OR}-\boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{m}_{R,Ex}\end{bmatrix},$$

where

$$m=\sum_{k=1}^{N}m_k,\boldsymbol{l}_{RC}=\frac{\sum_{k=1}^{N}m_k\boldsymbol{l}_{RP}}{m},\boldsymbol{f}_{Ex}=\sum_{k=1}^{N}\boldsymbol{f}_{Ex,k},\boldsymbol{m}_{R,Ex}=\sum_{k=1}^{N}\tilde{\boldsymbol{r}}_{RP}\boldsymbol{f}_{Ex,k},$$

$$\boldsymbol{J}_{R|R}=\sum_{k=1}^{N}m_k\tilde{\boldsymbol{l}}_{RP}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}=\sum_{k=1}^{N}m_k\left(\boldsymbol{l}_{RP}^2\boldsymbol{I}_3-\boldsymbol{l}_{RP}\boldsymbol{l}_{RP}^{\mathrm{T}}\right).$$

刚体元件动力学方程

刚体元件运动微分方程

$$\mathbf{0}=\begin{bmatrix}\sum_k(m_k\ddot{\boldsymbol{r}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP}-\boldsymbol{f}_{Ex,k})\\ \sum_k\tilde{\boldsymbol{l}}_{RP}\boldsymbol{A}_{OR}^{\mathrm{T}}(m_k\ddot{\boldsymbol{r}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RP}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR}+m_k\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RP}-\boldsymbol{f}_{Ex,k})\end{bmatrix},$$

可进一步化简为

$$\mathbf{0}=\begin{bmatrix}m\ddot{\boldsymbol{r}}_{OR}+m\boldsymbol{A}_{OR}\tilde{\boldsymbol{l}}_{RC}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR}+m\boldsymbol{A}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RC}-\boldsymbol{f}_{Ex}\\ m\tilde{\boldsymbol{l}}_{RC}\boldsymbol{A}_{OR}^{\mathrm{T}}\ddot{\boldsymbol{r}}_{OR}+\boldsymbol{J}_{R|R}\dot{\boldsymbol{\omega}}_{OR}+\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{J}_{R|R}\boldsymbol{\omega}_{OR}-\boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{m}_{R,Ex}\end{bmatrix},$$

动力学方程中的第一项左乘 $\boldsymbol{A}_{OR}^{\mathrm{T}}$ 后可得

$$\mathbf{0}=\begin{bmatrix}m\boldsymbol{A}_{OR}^{\mathrm{T}}\ddot{\boldsymbol{r}}_{OR}+m\tilde{\boldsymbol{l}}_{RC}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR}+m\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RC}-\boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{f}_{Ex}\\ m\tilde{\boldsymbol{l}}_{RC}\boldsymbol{A}_{OR}^{\mathrm{T}}\ddot{\boldsymbol{r}}_{OR}+\boldsymbol{J}_{R|R}\dot{\boldsymbol{\omega}}_{OR}+\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{J}_{R|R}\boldsymbol{\omega}_{OR}-\boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{m}_{R,Ex}\end{bmatrix}.$$

刚体元件动力学方程的矩阵形式

刚体元件动力学方程：

$$\mathbf{0} = \begin{bmatrix} m\boldsymbol{A}_{OR}^{\mathrm{T}}\ddot{\boldsymbol{r}}_{OR} + m\tilde{\boldsymbol{l}}_{RC}^{\mathrm{T}}\dot{\boldsymbol{\omega}}_{OR} + m\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RC} - \boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{f}_{Ex} \\ m\tilde{\boldsymbol{l}}_{RC}\boldsymbol{A}_{OR}^{\mathrm{T}}\ddot{\boldsymbol{r}}_{OR} + \boldsymbol{J}_{R|R}\dot{\boldsymbol{\omega}}_{OR} + \tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{J}_{R|R}\boldsymbol{\omega}_{OR} - \boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{m}_{R,Ex} \end{bmatrix}.$$

上式可进一步记为如下矩阵形式

$$\begin{bmatrix} m\boldsymbol{I}_3 & m\tilde{\boldsymbol{l}}_{RC}^{\mathrm{T}} \\ m\tilde{\boldsymbol{l}}_{RC} & \boldsymbol{J}_{R|R} \end{bmatrix} \begin{bmatrix} \boldsymbol{A}_{OR}^{\mathrm{T}}\ddot{\boldsymbol{r}}_{OR} \\ \dot{\boldsymbol{\omega}}_{OR} \end{bmatrix} + \begin{bmatrix} m\tilde{\boldsymbol{\omega}}_{OR}\tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{l}_{RC} \\ \tilde{\boldsymbol{\omega}}_{OR}\boldsymbol{J}_{R|R}\boldsymbol{\omega}_{OR} \end{bmatrix} = \begin{bmatrix} \boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{f}_{Ex} \\ \boldsymbol{A}_{OR}^{\mathrm{T}}\boldsymbol{m}_{R,Ex} \end{bmatrix},$$

并可简记为

$$\boldsymbol{M}_R\boldsymbol{a}_R - \boldsymbol{\beta}_R = \boldsymbol{f}_{R,Ex}.$$